Discrete Mathematics: Set Theory Notes

Module 2

1. Basic Set Operations

- **Union ($A \cup B$):** Combines elements from two sets.

$$A \cup B = \{x \mid (x \in A) \vee (x \in B)\}$$

- **Intersection ($A \cap B$):** Finds elements shared by both sets.

$$A \cap B = \{x \mid (x \in A) \wedge (x \in B)\}$$

- **Complement (A' or A^c):** Contains all elements not in A, relative to a universal set U.

$$A' = U - A = \{x \in U \mid x \notin A\}$$

- **Difference ($A - B$):** Contains elements in A that are not in B.

$$A - B = \{x \in A \mid x \notin B\}$$

2. Fundamental Laws of the Algebra of Sets

The Algebra of Sets obeys several core identity properties:

Commutative Laws

Order does not matter for union or intersection.

$$A \cup B = B \cup A$$
$$A \cap B = B \cap A$$

Associative Laws

Grouping does not matter when operations are the same.

$$(A \cup B) \cup C = A \cup (B \cup C)$$
$$(A \cap B) \cap C = A \cap (B \cap C)$$

Distributive Laws

Operations distribute over one another.

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$
$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Idempotent Laws

Combining a set with itself yields the same set.

$$A \cup A = A$$
$$A \cap A = A$$

De Morgan's Laws

Complements invert unions and intersections.

$$(A \cup B)' = A' \cap B'$$
$$(A \cap B)' = A' \cup B'$$

Identity Laws

Interactions with the universal set U or empty set $\emptyset$.

$$A \cup \emptyset = A$$
$$A \cap U = A$$